

Andrew Su

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EDUCATION

Columbia University

M.S. Applied Physics

New York City, NY

Aug 2026 – May 2027

University of California, Los Angeles

B.S. Physics, B.S. Computer Science

Los Angeles, CA

Sep 2022 – Jun 2026

- Overall GPA: 3.78/4.00, Dean's Honor List
- Upper Division Physics GPA: 3.95/4.00

RESEARCH

Columbia University, Department of Applied Physics & Applied Mathematics

New York City, NY

Graduate Student Researcher, Advisor: Professor Xueyue (Sherry) Zhang

Jul 2026 – Present

- Investigated waveguide-mediated long-range gates with tunable-coupling qubits (TCQs) for superconducting qubit architectures
- Modeled decoherence channels in TCQs analytically and developed numerical simulations to identify dominant loss mechanisms and improve qubit coherence
- Developed lab monitoring software to unify instrument telemetry, log status data, generate daily reports, and send Slack alerts for equipment malfunctions
- Designed and fabricated a custom electrostatic-dissipative acrylic SMA breakout board using Autodesk Fusion 360 and a Boss LS-3655 laser cutter

Brookhaven National Laboratory (BNL)

Upton, NY

SURP Research Intern, Advisor: Dr. Layla Hormozi

Aug 2025 – Jun 2026

- Extended prior summer research through BNL's Supplemental Undergraduate Research Program (SURP)
- Researched anyon errors, methods for improved vertex correction, and logarithmic time ground state initialization

Brookhaven National Laboratory (BNL)

Upton, NY

SULI Research Intern, Advisor: Dr. Layla Hormozi

Jun 2025 – Aug 2025

- Conducted theoretical research on non-Abelian topological quantum error-correcting codes through the Department of Energy's Science Undergraduate Laboratory Internship (SULI) Program
- Designed a topological quantum error-correcting architecture based on the string-net implementation of the doubled Fibonacci model, enabling scalable simulation and exploration of experimental feasibility
- Developed a framework in Python linking trivalent lattice geometry with quantum backends using Qiskit, quimb, and Matrix Product State (MPS) tensor network methods, providing an extendable platform for further study of the Fibonacci model

UCLA, Department of Physics & Astronomy

Los Angeles, CA

Undergraduate Researcher, Advisor: Professor Jay Hauser

Sep 2023 – Jun 2026

- Developed novel VR displays for particle collision events at the Compact Muon Solenoid (CMS) Experiment at the Large Hadron Collider with C# and Unity, granting researchers previously unattainable insight into complex aspects of collision data
- Expanded and optimized a parser for IG file conversion utilizing C# and Bash, ensuring seamless integration with Meta Quest headsets to display particle collider data in VR

UCLA Nuclear Physics, Department of Physics & Astronomy

Los Angeles, CA

Undergraduate Researcher, Advisor: Professor Huan Huang

Oct 2023 – Sep 2024

- Performed analysis using ROOT, C++, and Bash on data from the Relativistic Heavy Ion Collider at Brookhaven National Laboratory, extracting Fourier coefficients to study anisotropic flow in relativistic nuclear collisions

CLUBS & ASSOCIATIONS

Sigma Pi Sigma Physics Honor Society — President

Apr 2024 – Mar 2026

- Raised \$50,000 and founded an undergraduate research fellowship program supporting students with funded research opportunities
- Organized and raised over \$3,000 for the inaugural transfer summit to support community college transfers into physics, involving over a dozen faculty from various schools across Southern California, and with over 200 student participants
- Orchestrated membership growth and recruitment initiatives, increasing chapter size by 200% through targeted outreach and value-added offerings
- Instituted a comprehensive academic document archive, physics review sessions, and a paired mentorship system, providing significant academic and professional benefits to members

William F. Sharpe Fellowship

- Selected as one of 14 fellows for the UCLA Sharpe Fellowship in Economics Tech Cohort, recognizing excellence in leadership, academic achievement, and civic engagement